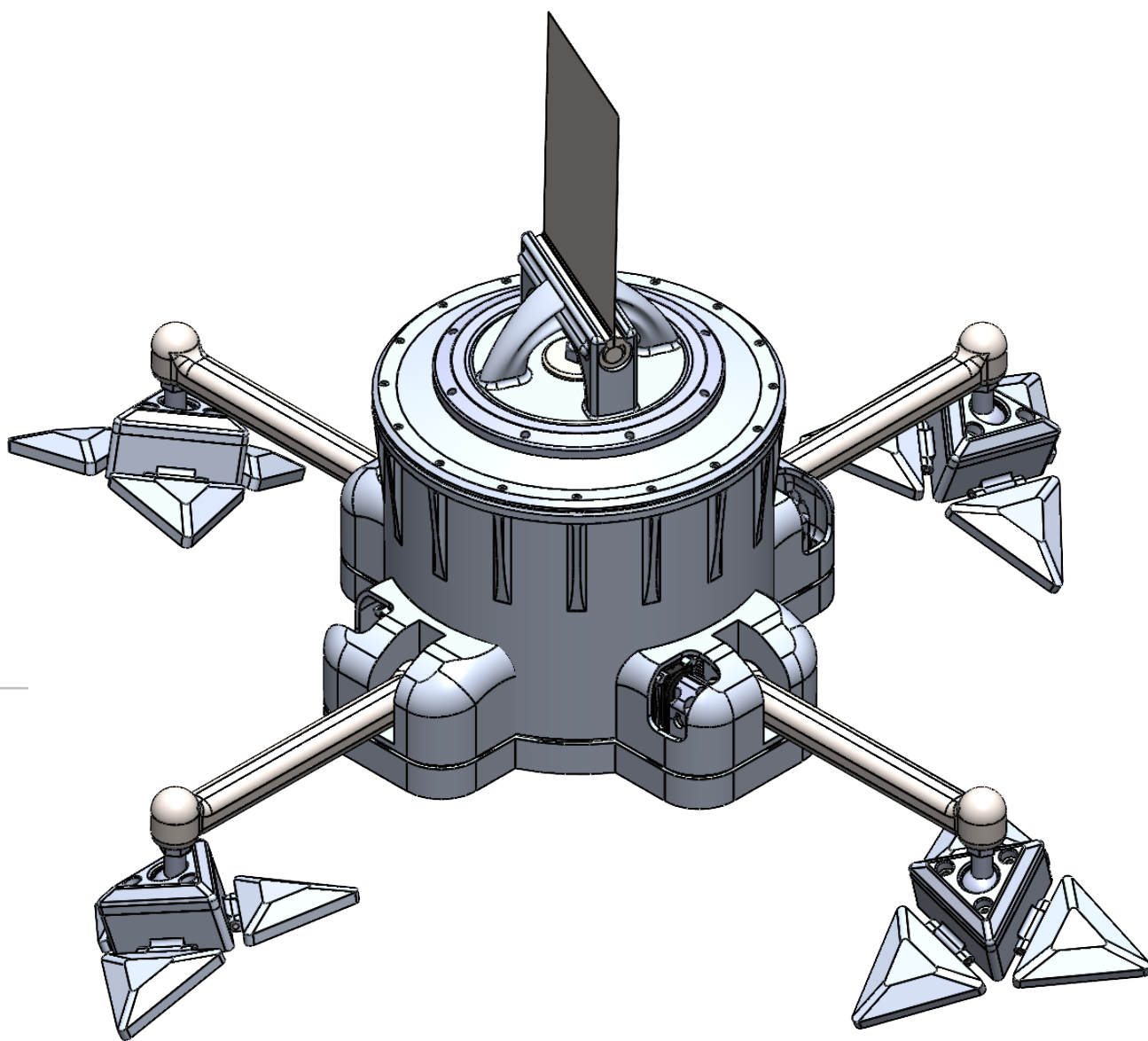




UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND
BREAK SHARP
EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE		
DRAWN	Jarek Shahim		6/9/25		
CHK'D					
APPV'D					
MFG					
Q.A					

TITLE:

Full Assembly

DWG NO.

V7

A4

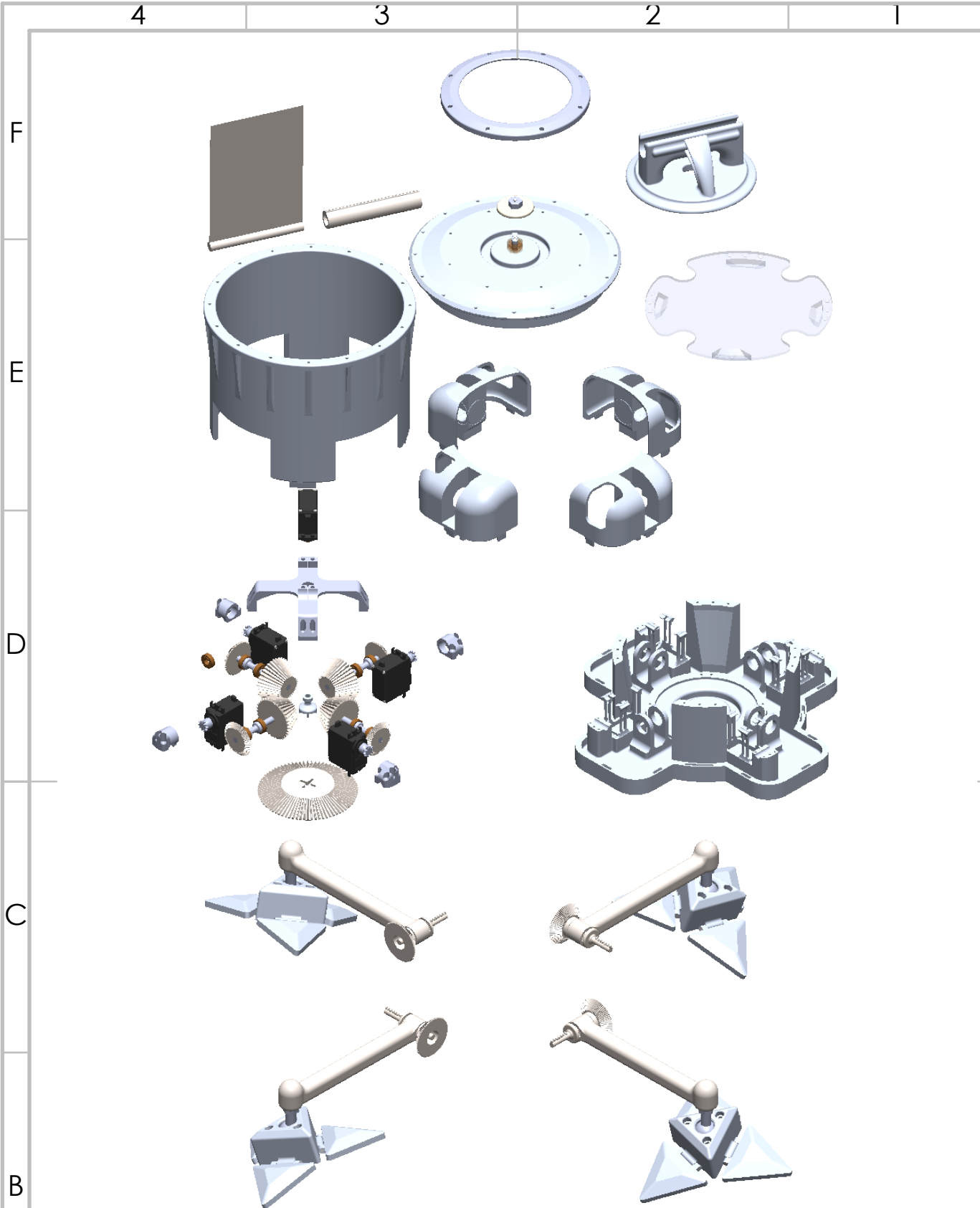
MATERIAL:

WEIGHT:

SCALE:1:3.5

SHEET 1 OF 1

4				3				2				1											
ITEM NO.		PART NUMBER				DESCRIPTION				QTY.													
F	1	Spur_Rack Key				Drives leg shaft lock				4													
	2	MG996R Motor								5													
	3	Base Plate				Main sturcture for gear train				1													
	6	bearing								11													
	7	Leg Cap				Acts as both housing the main structure and as leg support needs further refinement				4													
E	8	mounting plate								1													
	9	Main Motor Bracket								1													
	10	Main Motor Drive Shaft								1													
	11	Inch - Straight bevel pinion 20DP80PT 35GT 20PA 1FW ---80O4H.25MD0.125N				Secondary driving gear meshes with main gear, Sourceable				1													
	12	Drive Shaft 1/4" D x 3"				Main drive shaft for gear train, Sourceable				4													
D	13	Main Drive Gear - Inch - Straight bevel pinion 20DP35PT 80GT 20PA 1FW ---35O4H.25MD0.125N				Main driving gear for all four legs, special features - centering cavity and bearing cavity				4													
	14	Inch - Straight bevel pinion 48DP70PT 70GT 20PA .25FW ---70O.25H1MD0.25N				Tertiary gear meshes with lactching drive gear, Sourceable				4													
	15	Housing Top Plate				Main fixture point for gimbal				1													
	16	Gimbal Attachment ring				Constrains Gimbal				1													
	17	Swivel Base				Main structure of gimbal assmebly				1													
C	18	Tether Attachment Pin				locks tether to gimbal				1													
	19	Black Box				Main Structual shell				1													
	20	Thether				Mock up of space thether for attaching to gimbal				1													
	21	Swivel Washer				Welded to top plate, fixes gimbal assembly, Sourceable				1													
	22	.5" nut								1													
B	23	Mock Leg				Temporary place holder for leg Assembly				4													
	24	Ball-ankle								4													
	25	Foot								12													
	26	Ball_Hinge_top_plate								4													
	27	Ball_Hinge_Bottom_plate								4													
A	28	Leg_pin								12													
	29	Inch - Straight bevel pinion 40DP70PT 70GT 20PA 0.5FW ---70O.25H1MD0.25N								4													
	30	Inch - Rack-spur - rectangular 24DP 20PA .1FW .25PH 0.875L---SAll								4													
	31	SupportCapV3								4													
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:				FINISH:				DEBURR AND BREAK SHARP EDGES				DO NOT SCALE DRAWING				REVISION							
A	NAME		SIGNATURE		DATE								TITLE:		CAD Parts List BOM								
	DRAWN		Jarell Shahim		6/9/25																		
	CHK'D																						
	APPV'D																						
	MFG																						
								MATERIAL:				DWG NO.				Full Assembly BOM				A4			
												SCALE:1:20				SHEET 1 OF 1							
								WEIGHT:															



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND
BREAK SHARP
EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE		
DRAWN	Jareh Shahim		6/9/25		
CHK'D					
APPV'D					
MFG					
Q.A					

MATERIAL: Aluminum 6061
Titanium 5.2Sn

WEIGHT:

TITLE:

Exploded View

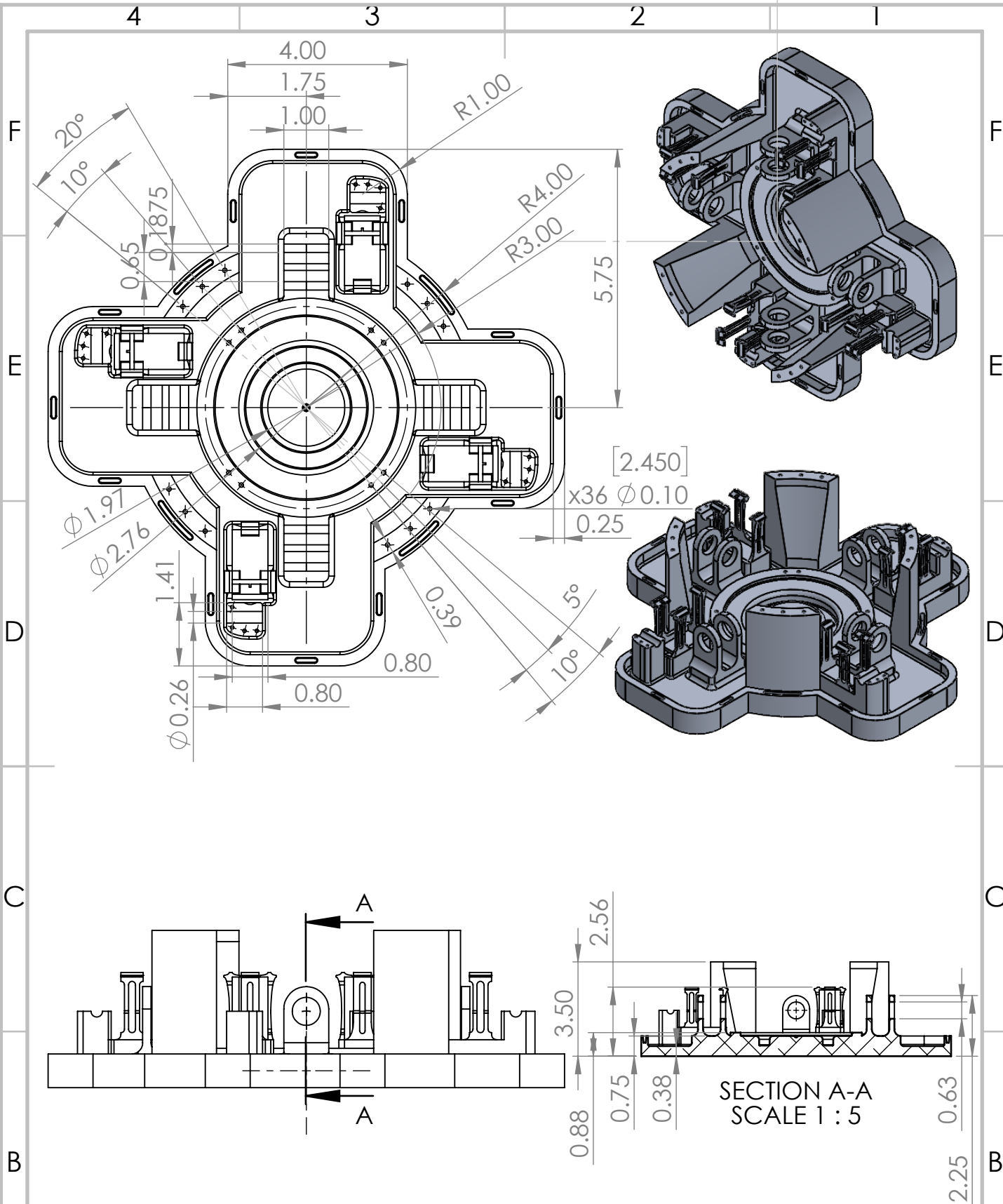
DWG NO.

Full Assembly

A4

SCALE:1:5.5

SHEET 1 OF 1



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND
BREAK SHARP
EDGES

DO NOT SCALE DRAWING

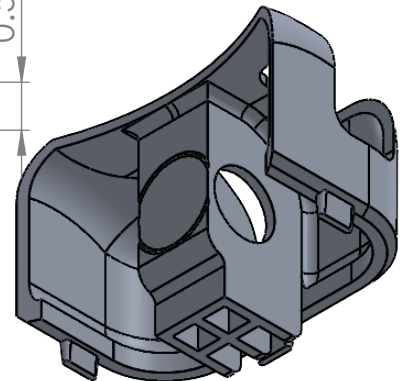
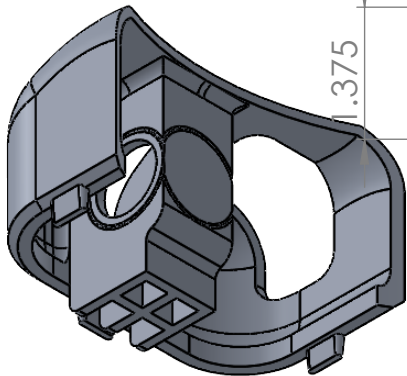
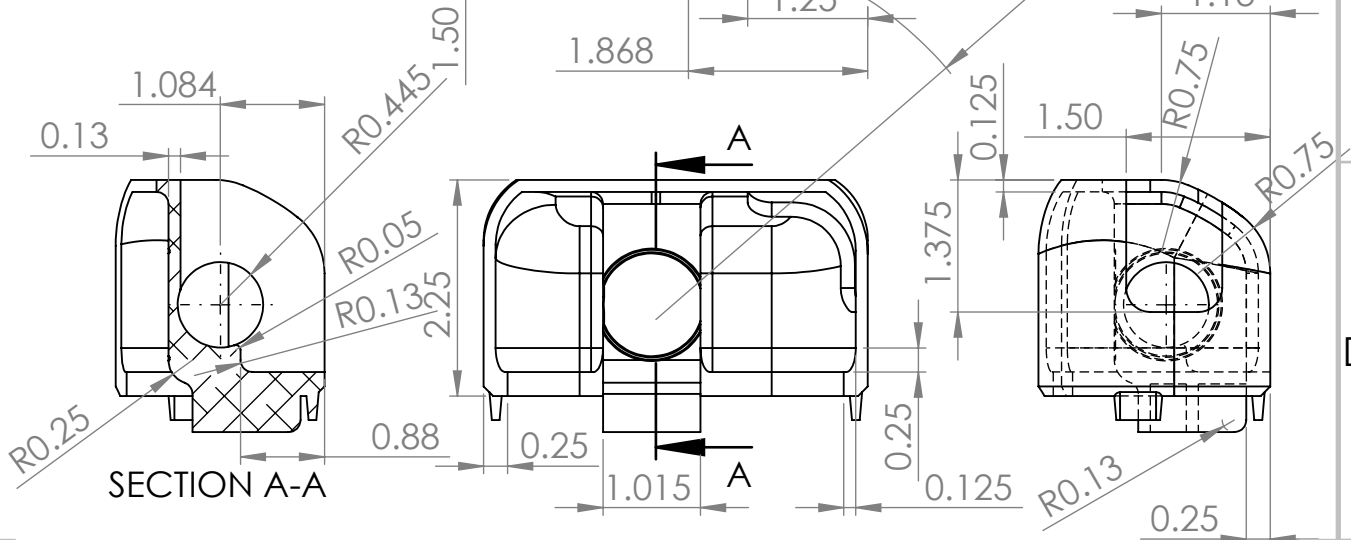
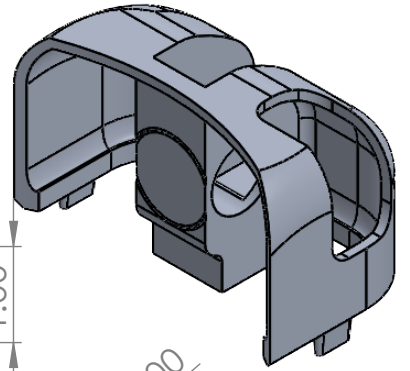
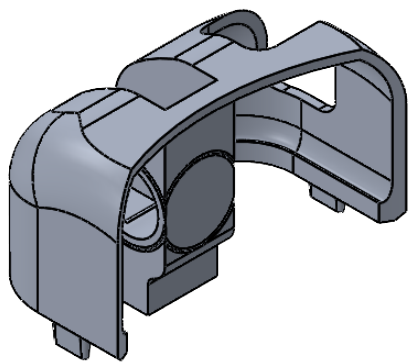
REVISION

	NAME	SIGNATURE	DATE
DRAWN	Jarel Shahim		06/09/25
CHK'D			
APPV'D			
MFG			
Q.A			

MATERIAL:	Aluminum 6061
WEIGHT:	

TITLE:	Main sturcture for gear train
DWG NO.	Base_Plate_V4
SCALE:1:3	SHEET 1 OF 1

A4



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND
BREAK SHARP
EDGES

DO NOT SCALE DRAWING

REVISION

NAME	SIGNATURE	DATE			
DRAWN Jareh Shahim		6/9/25			
CHK'D					
APPV'D					
MFG					
Q.A					
MATERIAL:			Aluminum 6061		
WEIGHT:					

TITLE:

Acts as both housing the main
structure and as leg support needs
further refinement

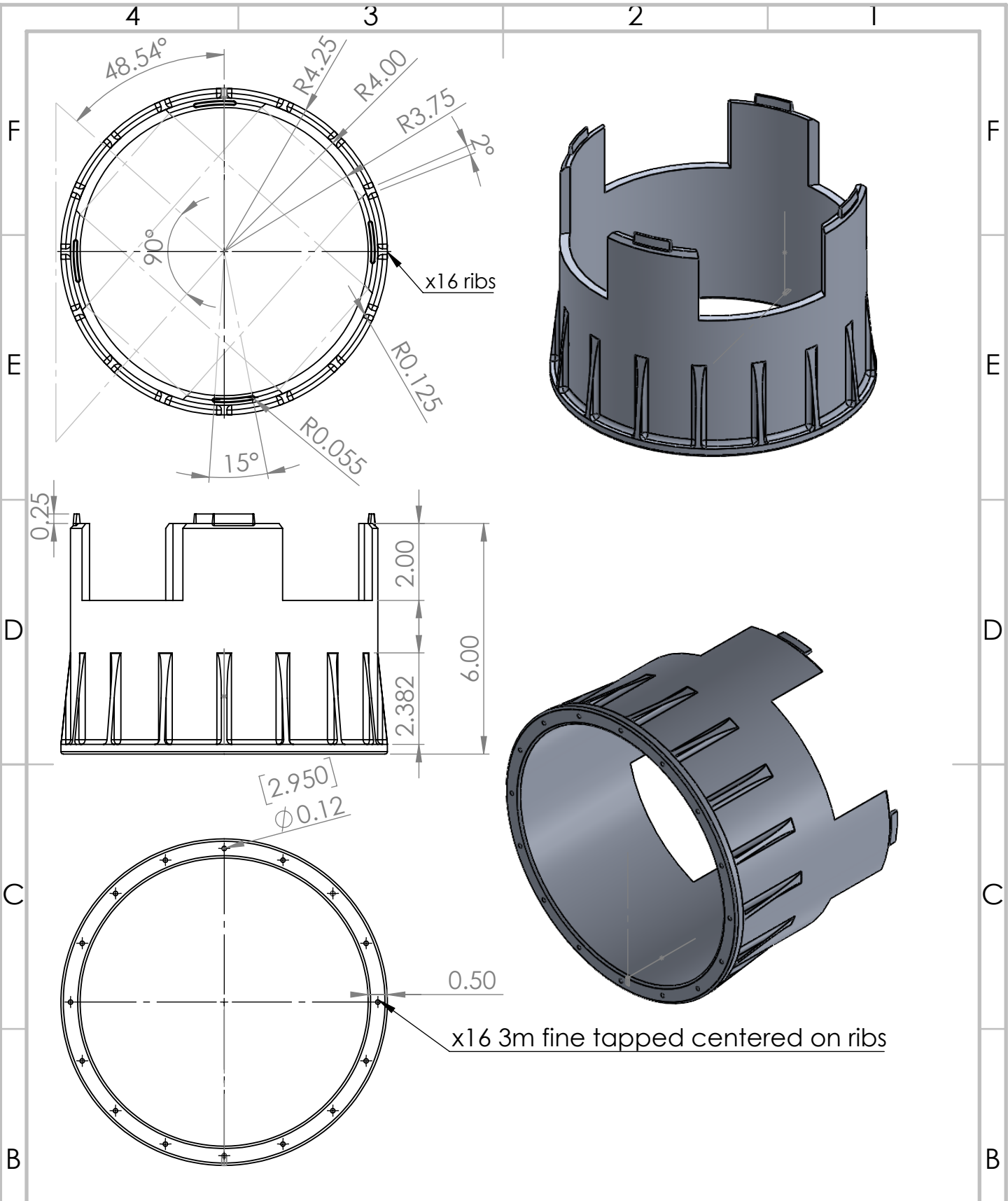
DWG NO.

Leg_Cap_V2

A4

SCALE:1:2

SHEET 1 OF 1



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND
BREAK SHARP
EDGES

DO NOT SCALE DRAWING

REVISION

NAME	SIGNATURE	DATE
DRAWN Jarell Shahim		06/08/2025
CHK'D		
APPV'D		
MFG		
Q.A		

MATERIAL:
Aluminum 6061

WEIGHT:

TITLE: Main structural shell, distributes load from gimbal to legs protects the internal mechanisms

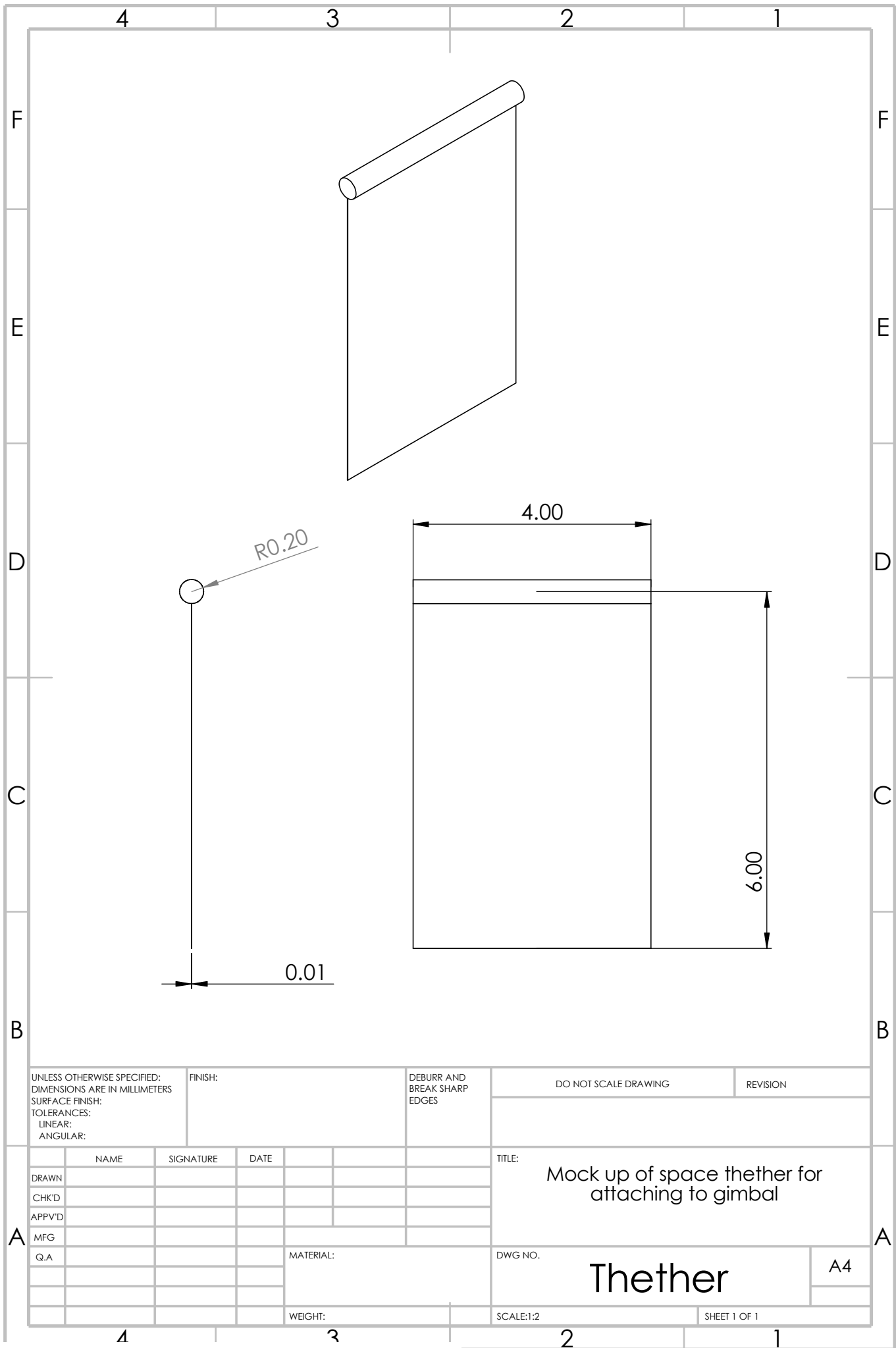
DWG NO.

BlackBox_V2

A4

SCALE:1:3.5

SHEET 1 OF 1



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND
BREAK SHARP
EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE			
DRAWN						
CHK'D						
APPV'D						
MFG						
Q.A						

TITLE:

Mock up of space tether for
attaching to gimbal

DWG NO.

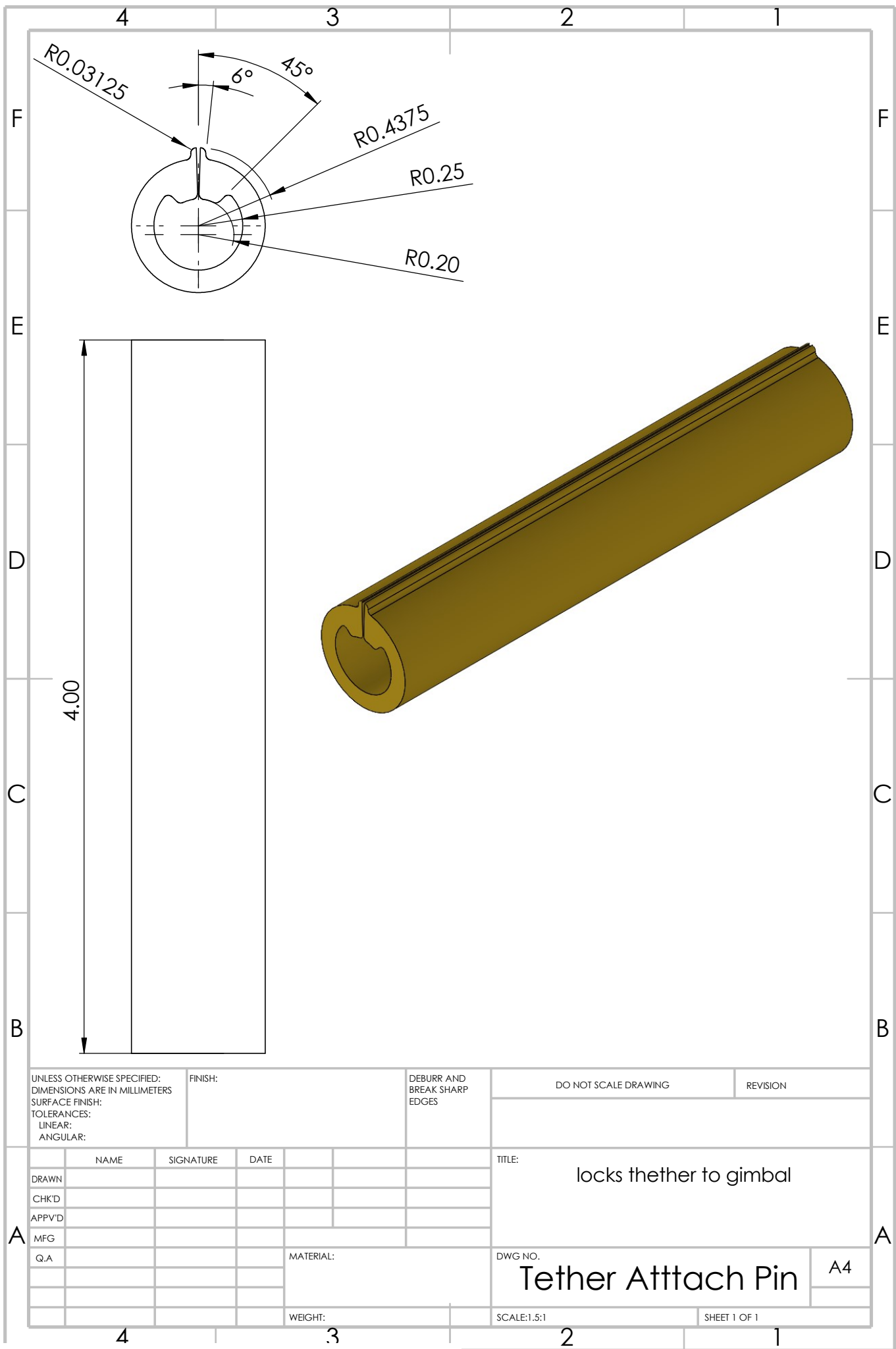
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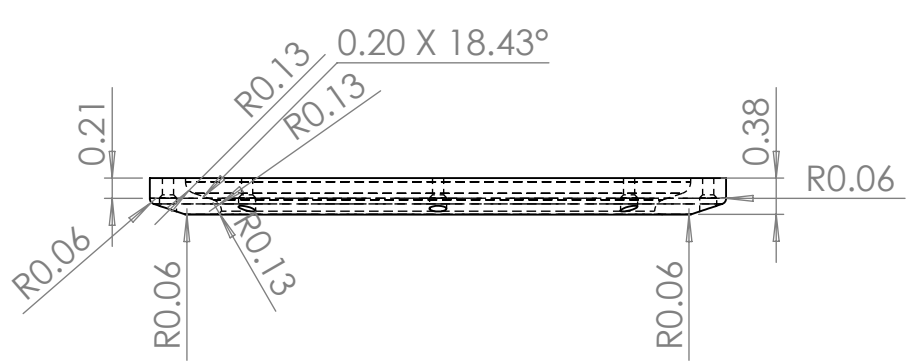
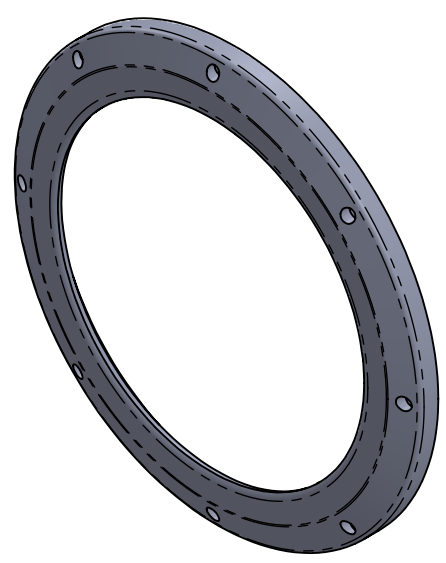
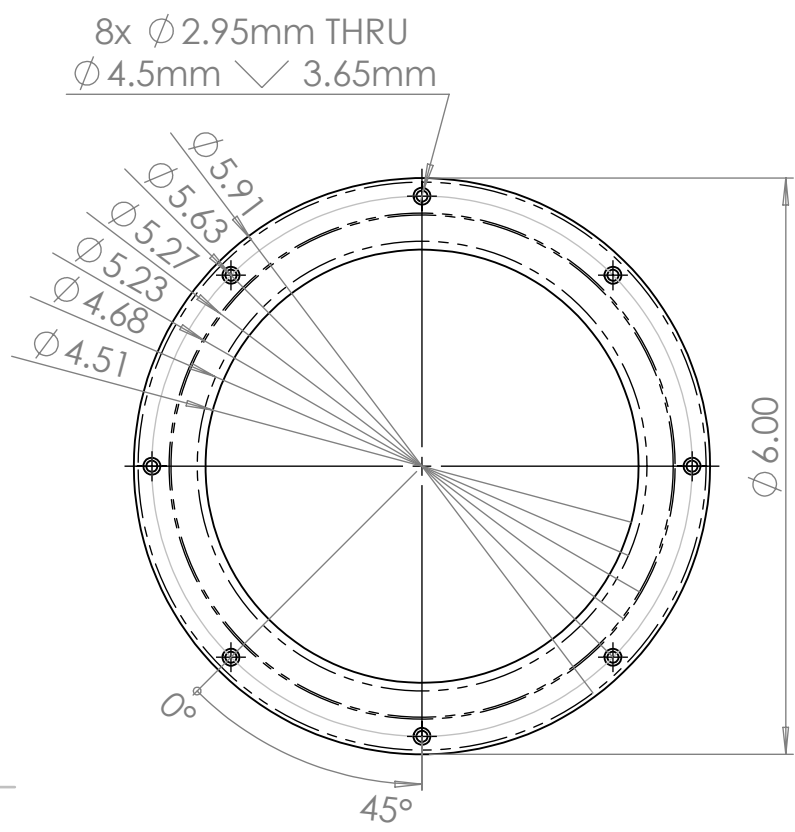
A4

WEIGHT:

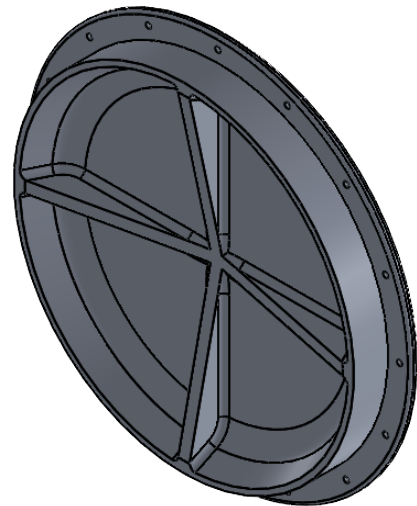
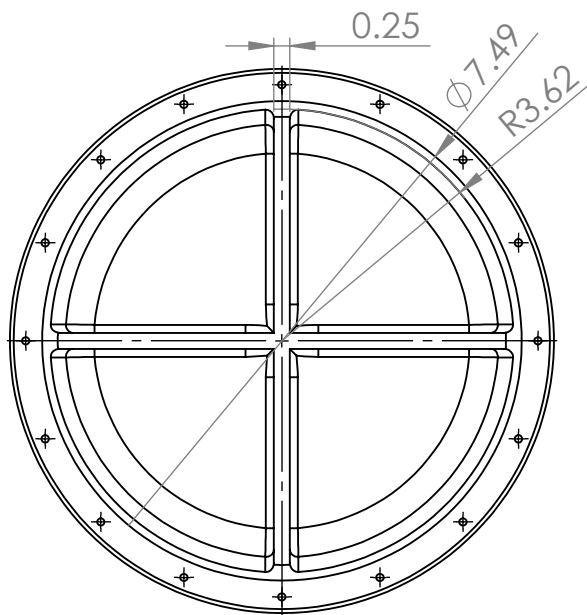
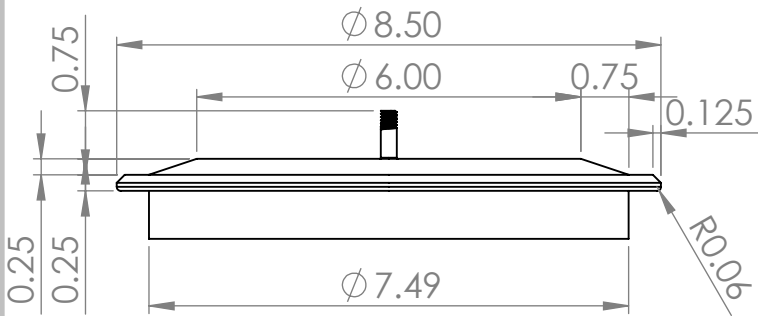
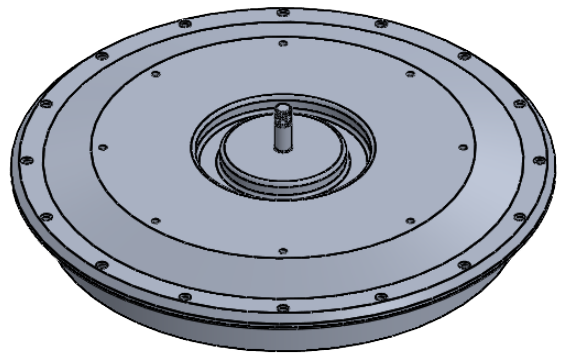
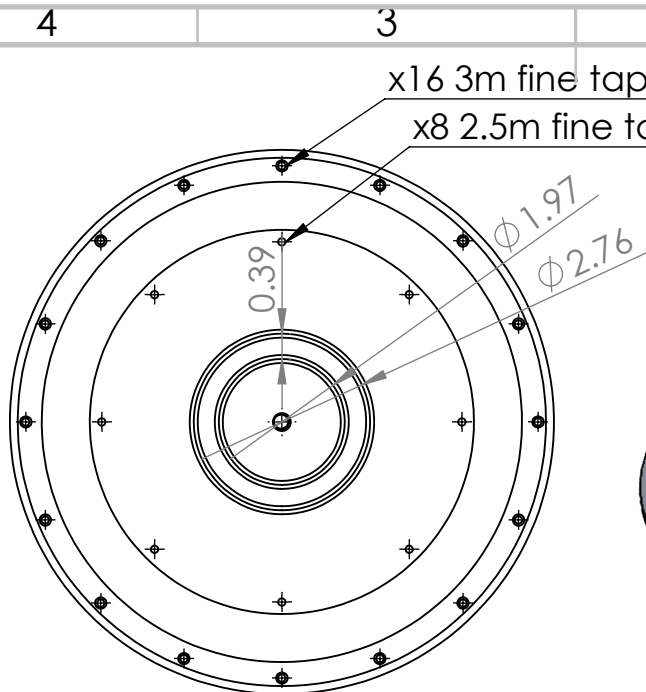
SCALE:1:2

SHEET 1 OF 1





UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:				FINISH:		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING		REVISION	
DRAWN				NAME		SIGNATURE		DATE		TITLE: Ring which holds the tether gimbal	
CHK'D											
APPV'D											
MFG											
Q.A											
								MATERIAL:		DWG NO. Gimbal Attachment ring	
								WEIGHT:		A4	
								SCALE:1:2		SHEET 1 OF 1	



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND
BREAK SHARP
EDGES

DO NOT SCALE DRAWING

REVISION

NAME	SIGNATURE	DATE		
DRAWN Jareh Shahim		06/09/25		
CHK'D				
APPV'D				
MFG				
Q.A				
MATERIAL:			Aluminum 6061	
WEIGHT:				

TITLE:

Main fixture point
for gimbal

DWG NO.

HousingTopPlate

A4

SCALE:1:3

SHEET 1 OF 1

4

3

2

1

F

F

E

E

D

D

C

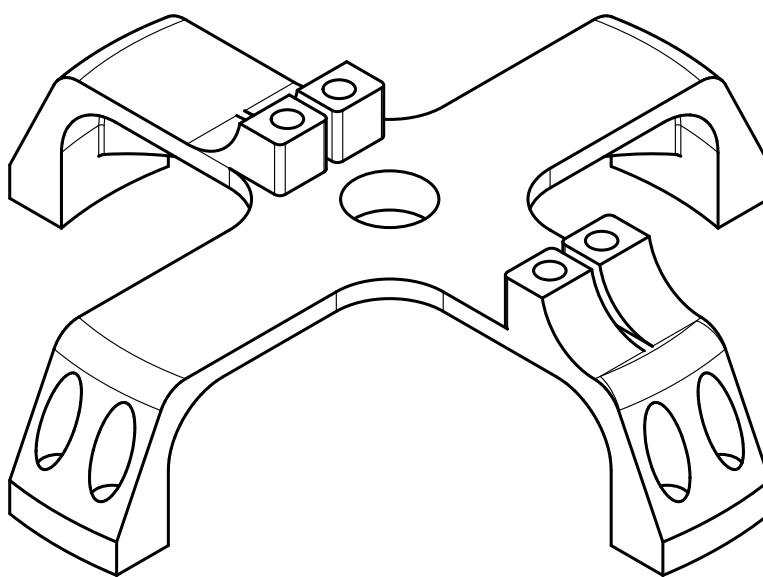
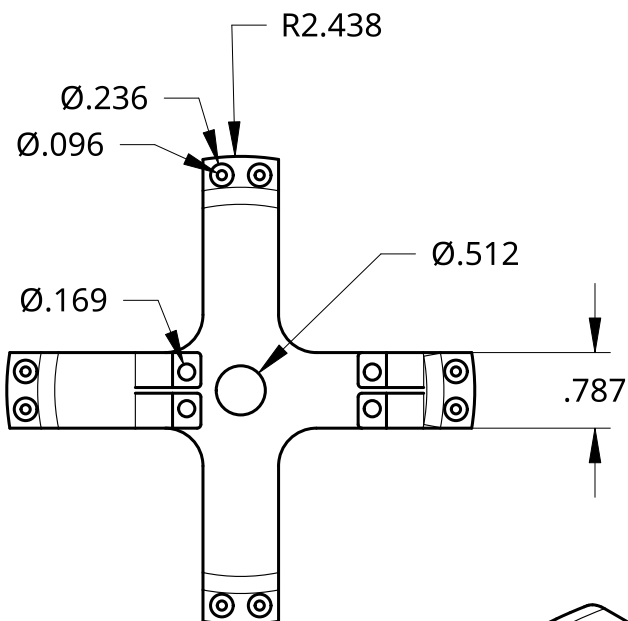
C

B

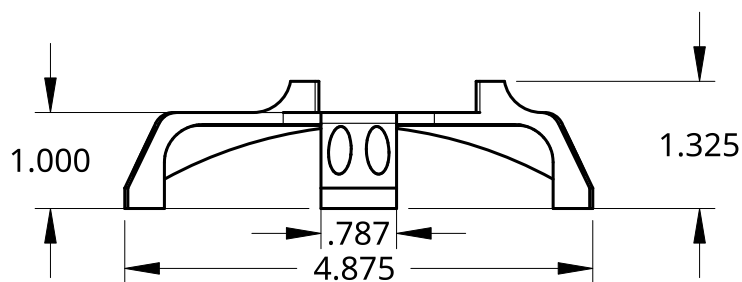
B

A

A



SCALE 1:1



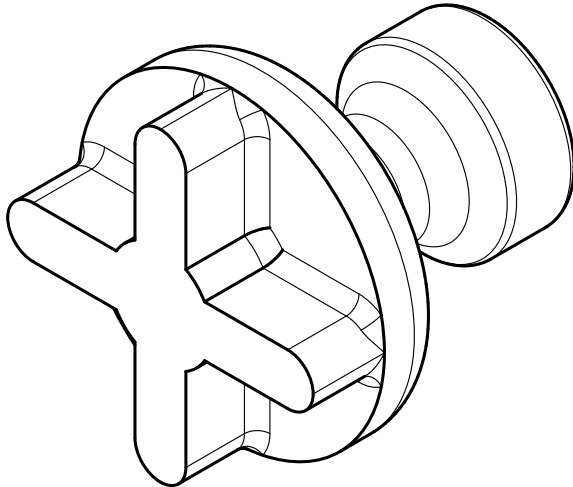
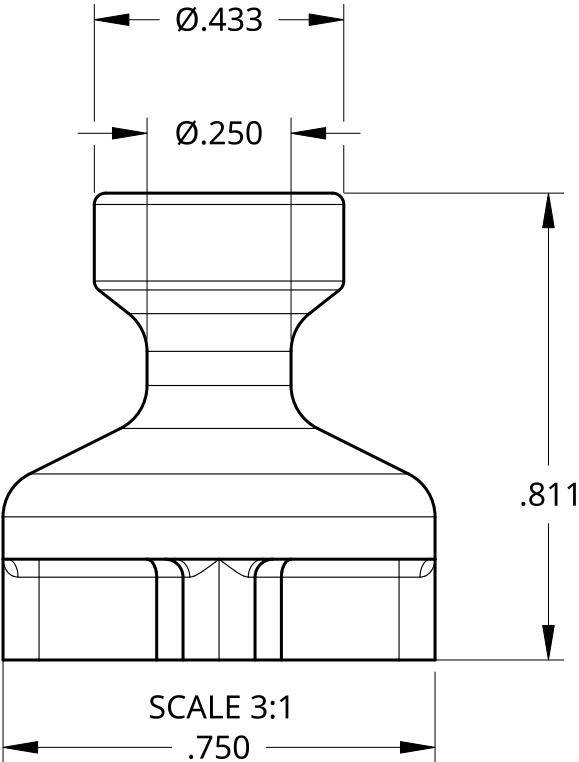
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES .XX = ±.0- .XXX = ±.00- .XXXX = ±.000- SURFACE FINISH 		NAME	DATE	TITLE Main Motor Bracket		
	DRAWN	KIEFER GONSALVES	06/10/2025			
	CHECKED					
	APPROVED					
	DO NOT SCALE DRAWING					
BREAK ALL SHARP EDGES AND REMOVE BURRS				SIZE A		
THIRD ANGLE PROJECTION 	MATERIAL Aluminum 6061	FINISH		DWG NO.	REV.	
				SCALE 1:2	WEIGHT	SHEET 1 of 1

4

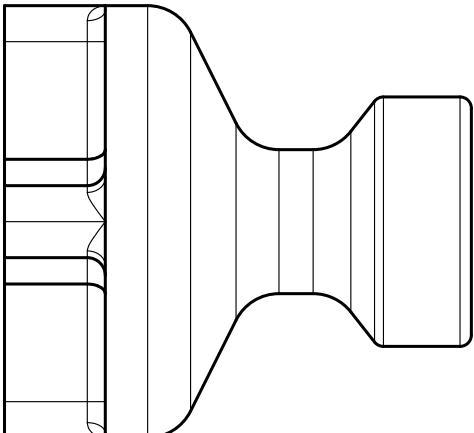
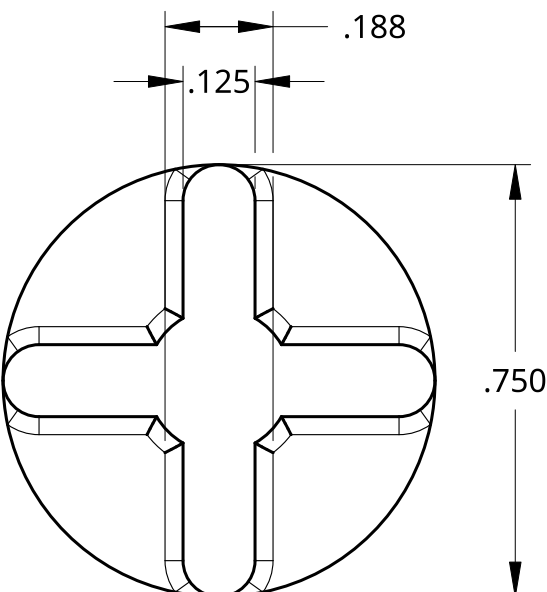
3

2

1



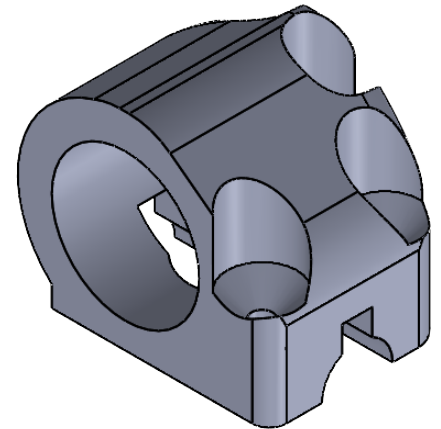
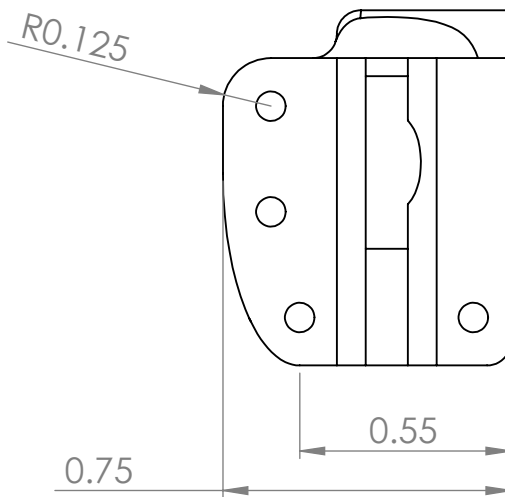
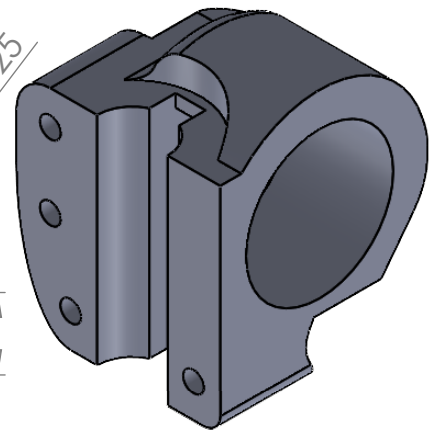
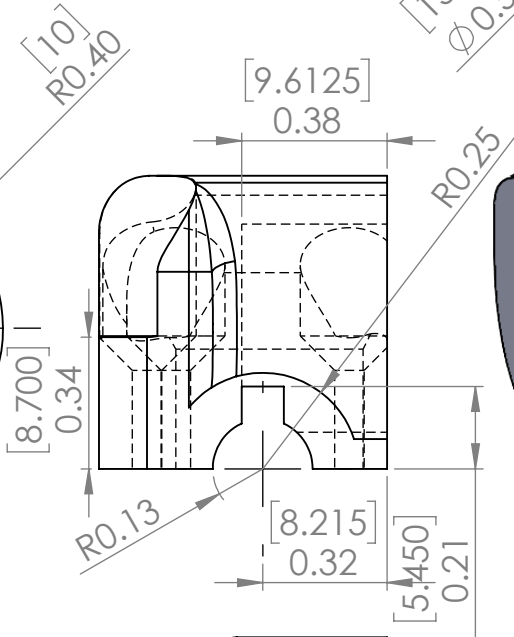
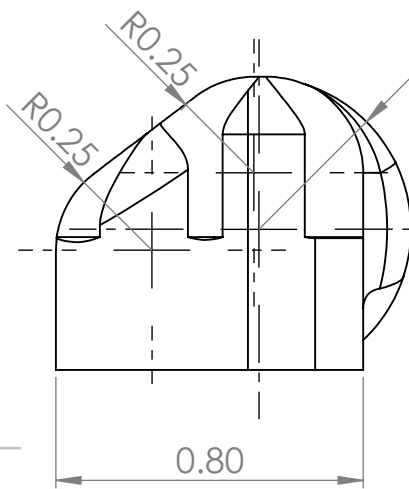
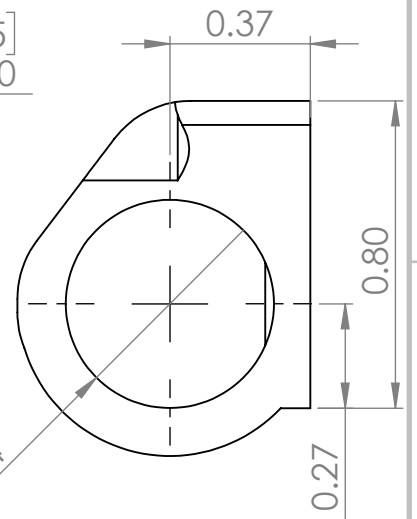
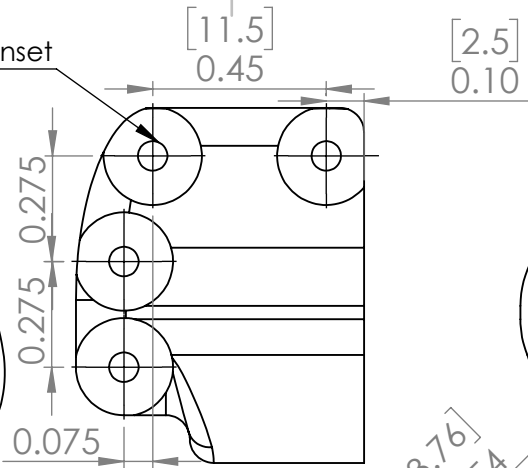
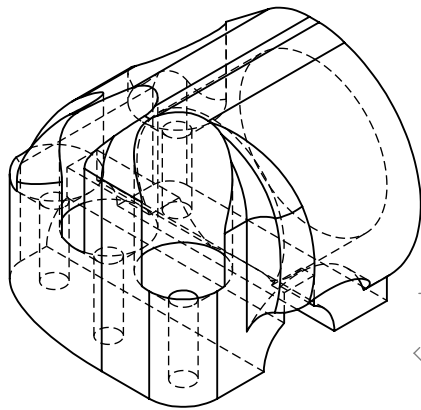
SCALE 3:1



SCALE 3:1

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES .XX = ±.0- .XXX = ±.00- .XXXX = ±.000- SURFACE FINISH 		NAME	DATE	TITLE Main Motor Drive Shaft		
	DRAWN	KIEFER GONSALVES	06/10/2025			
	CHECKED					
	APPROVED					
	DO NOT SCALE DRAWING					
BREAK ALL SHARP EDGES AND REMOVE BURRS				SIZE A		
THIRD ANGLE PROJECTION 	MATERIAL Titanium 5.2Sn	FINISH		DWG NO.		
				SCALE 2:1	WEIGHT	SHEET 1 of 1

x4 3m fine trapped with bevel inset



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND
BREAK SHARP
EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE		
DRAWN	Jarell Shahim		6/9/25		
CHK'D					
APPV'D					
MFG					
Q.A					
			MATERIAL:		
			Aluminum 6061		
			WEIGHT:		

TITLE:
Incloses and constrains leg lock shaft,
for motor drive gear.

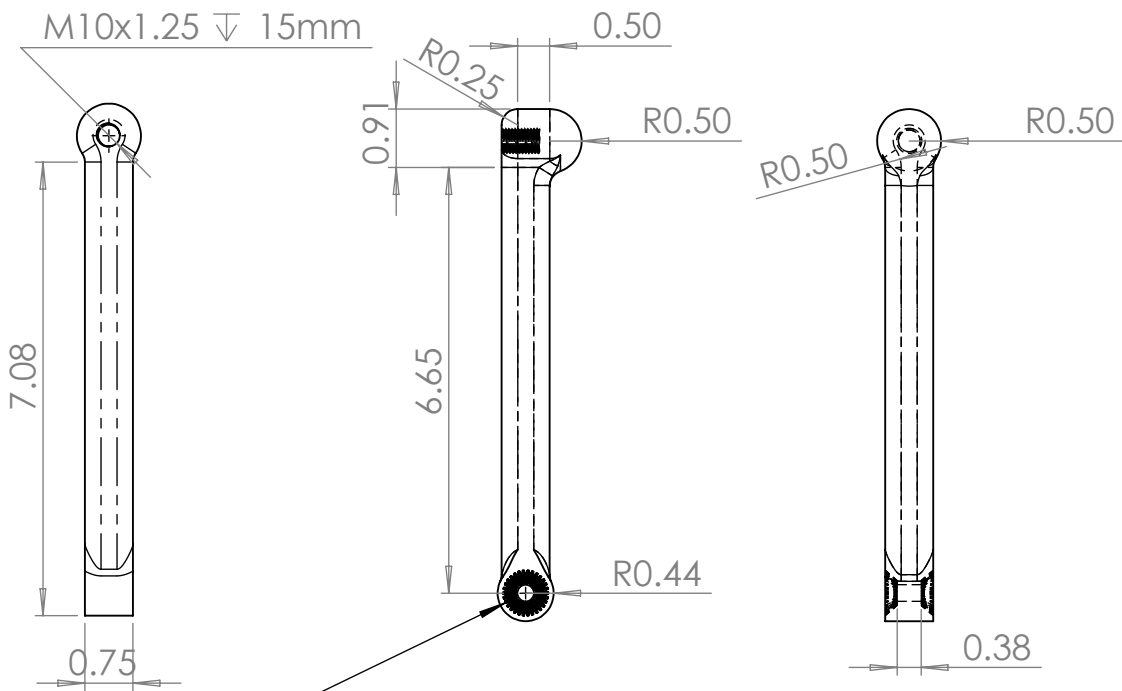
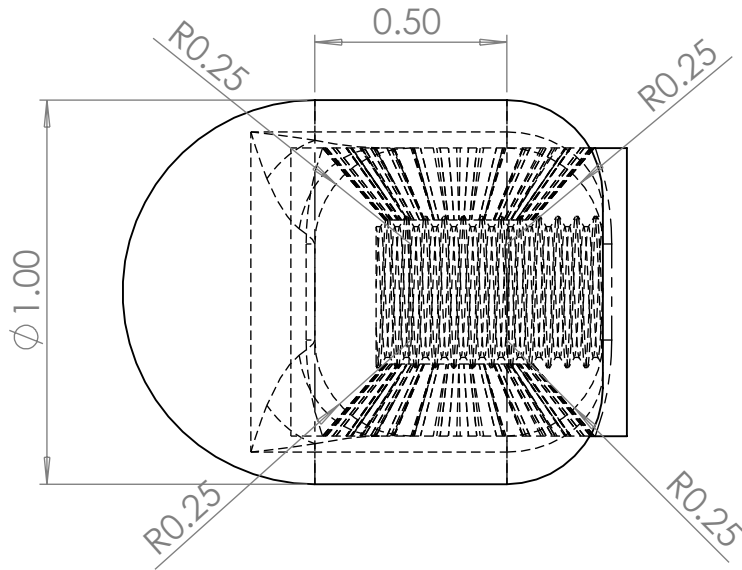
DWG NO.

SupportCapV3

A4

SCALE:2:1

SHEET 1 OF 1



24DP-20PA-1FW-0.25PH-0.875L

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND
BREAK SHARP
EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE		
DRAWN					
CHK'D					
APPV'D					
MFG					
Q.A					

MATERIAL:

WEIGHT:

TITLE:

Leg with bevel gear drive and
attachment for ball joint thread

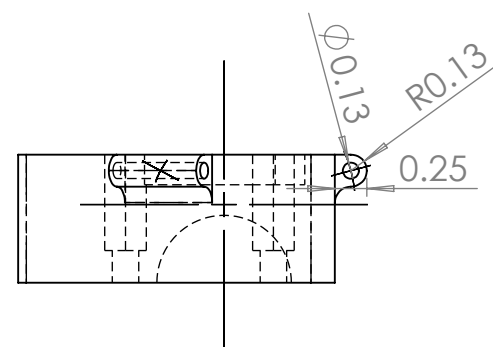
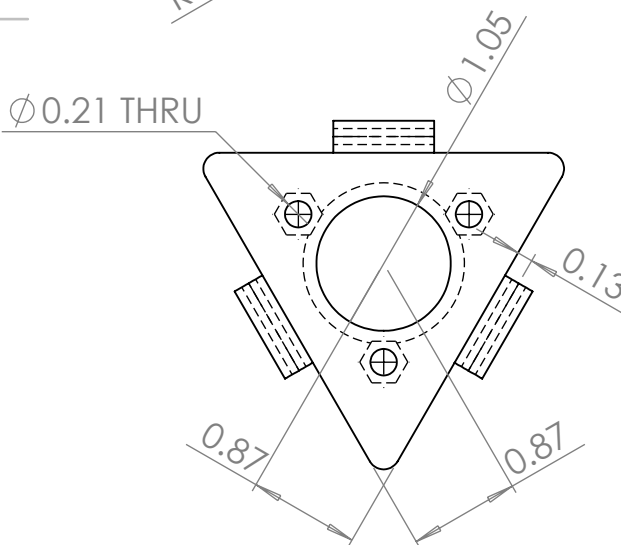
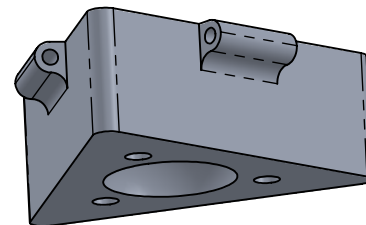
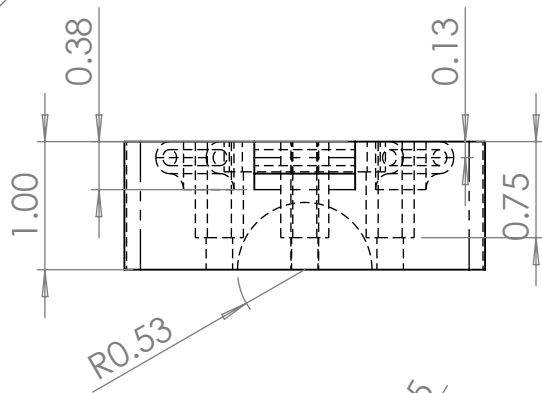
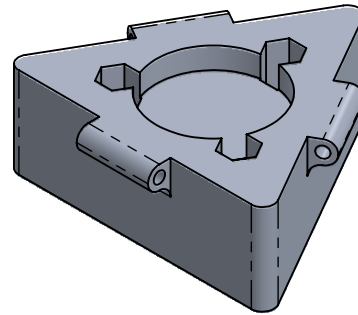
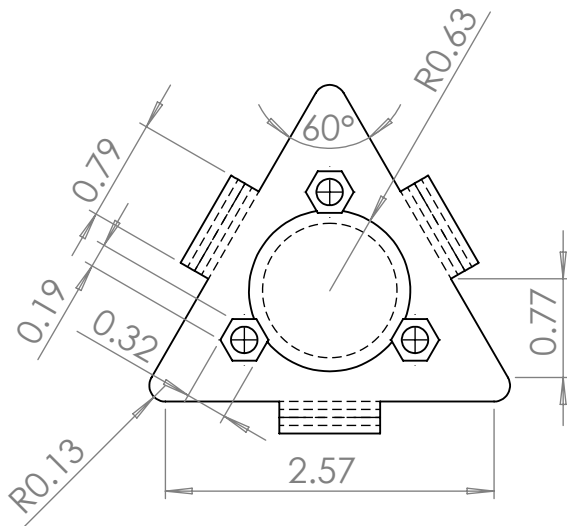
DWG NO.

mock_legv2

A4

SCALE:1:3

SHEET 1 OF 1



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND
BREAK SHARP
EDGES

DO NOT SCALE DRAWING

REVISION

NAME	SIGNATURE	DATE		
DRAWN				
CHK'D				
APPV'D				
MFG				
Q.A				

MATERIAL:

WEIGHT:

TITLE:

Bottom clamp for ball joint with press fit for magnet and three axels for feet.

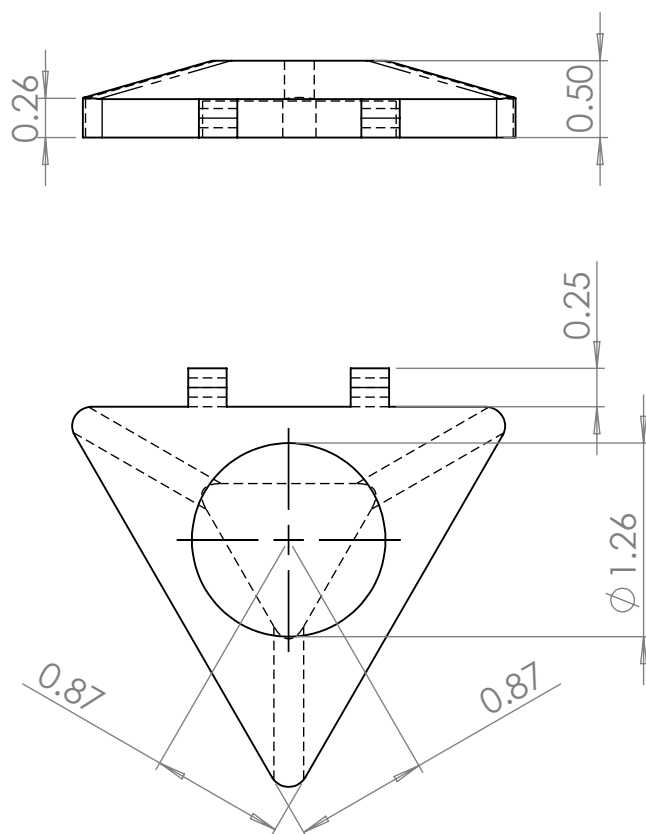
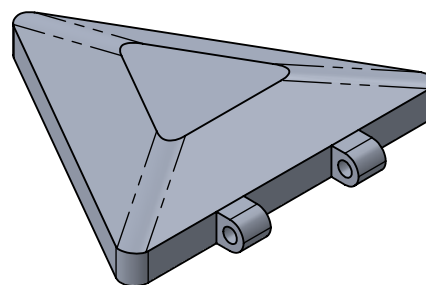
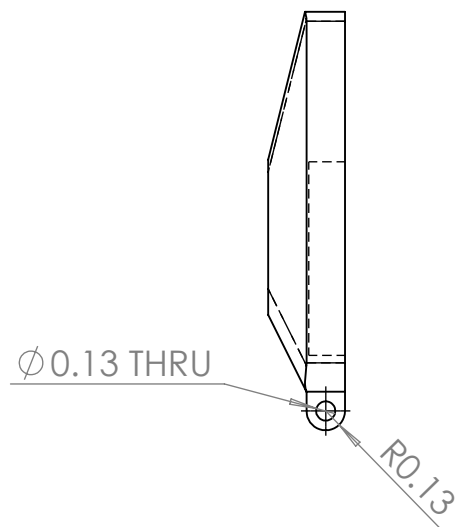
DWG NO.

Ball_Hinge_Bottom_plate

A4

SCALE:1:1.5

SHEET 1 OF 1



REVISION

SHEET 1 OF 1

4

3

2

1

F

F

E

E

D

D

C

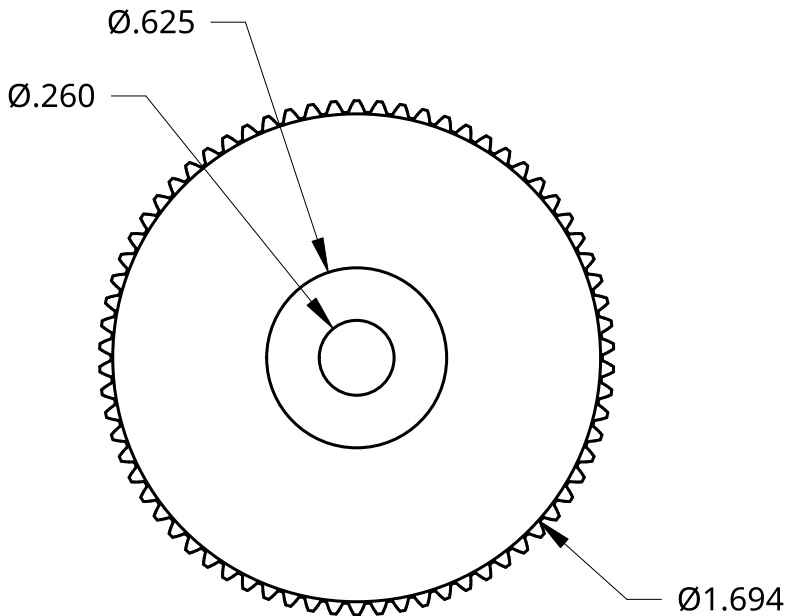
C

B

B

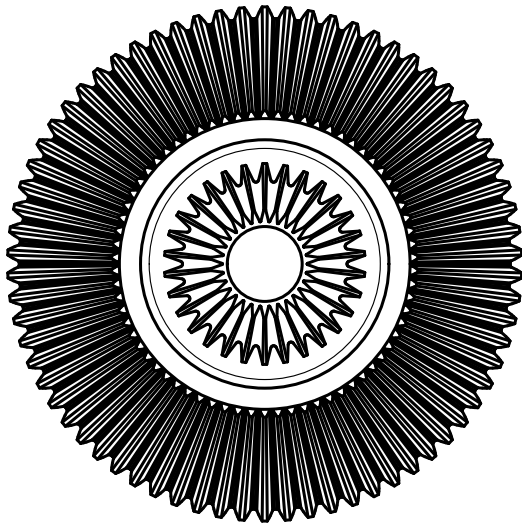
A

A

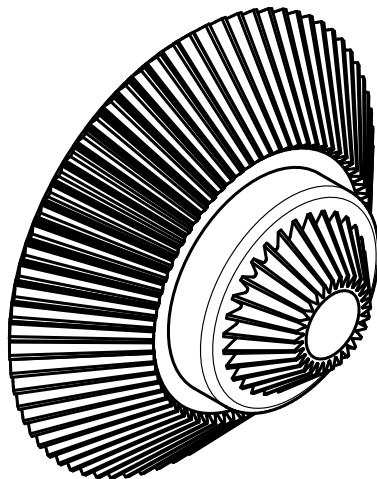


SCALE 3:2

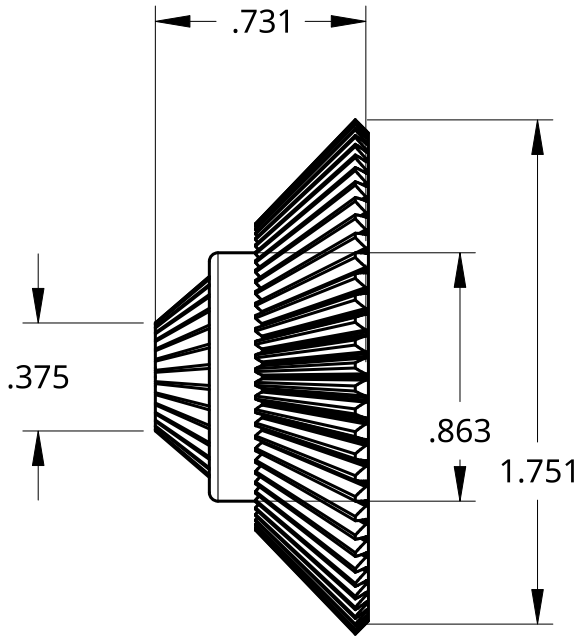
48DP 70PT 70GT 20PA .25FW



SCALE 3:2



SCALE 3:2



SCALE 3:2

UNLESS OTHERWISE SPECIFIED,
DIMENSIONS ARE IN INCHES

.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-

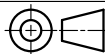
ANGULAR = ± °
FRACTIONAL = ±

SURFACE FINISH

DO NOT SCALE DRAWING

BREAK ALL SHARP EDGES AND
REMOVE BURRS

THIRD ANGLE PROJECTION



NAME	DATE
DRAWN	KIEFER GONSALVES
CHECKED	
APPROVED	
MATERIAL	FINISH
Titanium	
5.2Sn	

TITLE		
48DP70PT 70GT 20PA .25FW -V4		
SIZE	DWG NO.	REV.
A		
SCALE	WEIGHT	SHEET
1:1		1 of 1

4

3

2

1

F

F

E

E

D

D

C

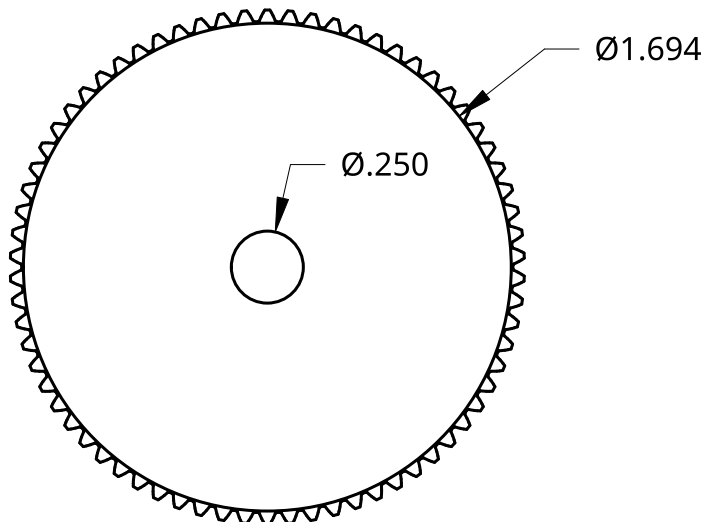
C

B

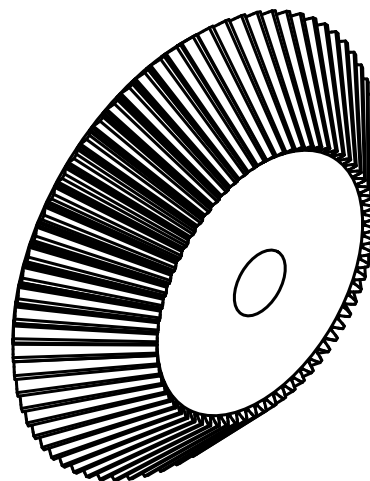
B

A

A

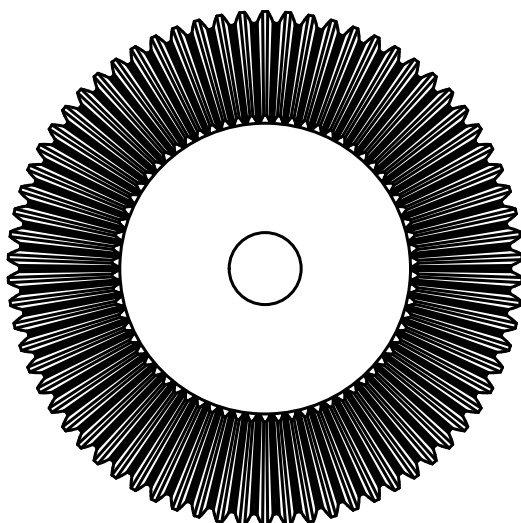


SCALE 3:2

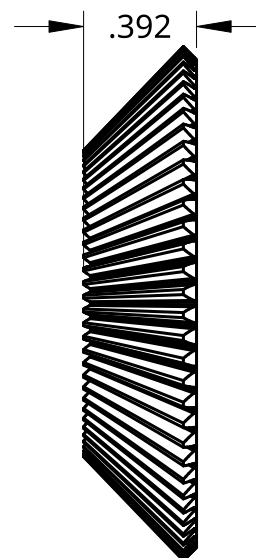


SCALE 3:2

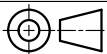
48DP 70PT 70GT 20PA .25FW



SCALE 3:2



SCALE 3:2

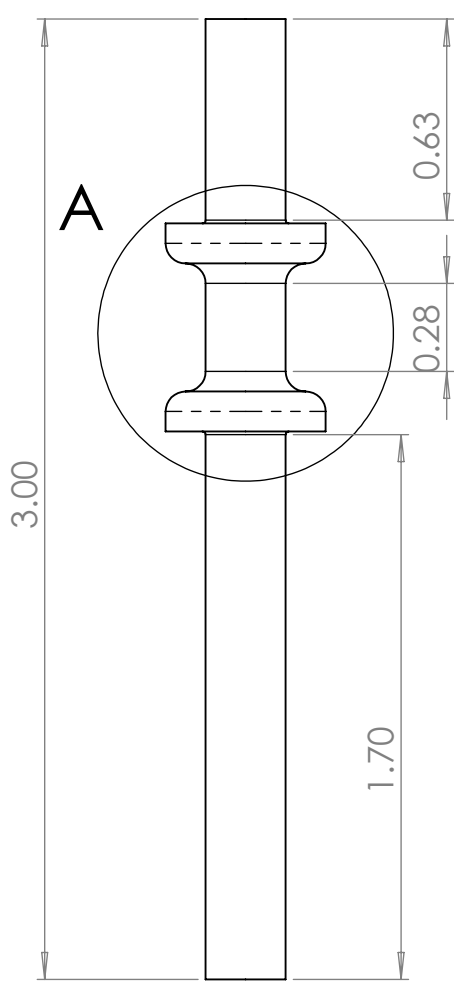
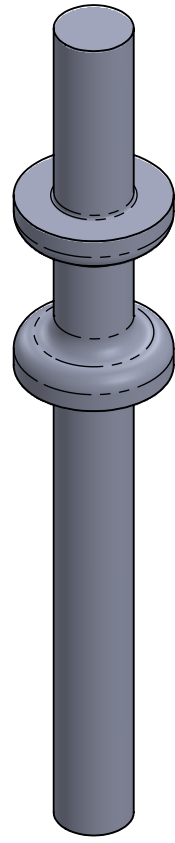
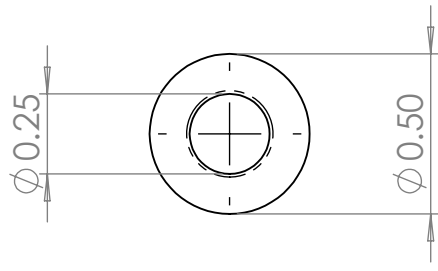
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES . XX = ±.0- .XXX = ±.00- .XXXX = ±.000- SURFACE FINISH  DO NOT SCALE DRAWING BREAK ALL SHARP EDGES AND REMOVE BURRS THIRD ANGLE PROJECTION 		NAME	DATE	TITLE Third-Straight bevel pinion V4		
	DRAWN	KIEFER GONSALVES	06/10/2025			
	CHECKED					
	APPROVED					
				SIZE A	DWG NO.	REV.
	MATERIAL Titanium 5.2Sn	FINISH		SCALE 1:1	WEIGHT	SHEET 1 of 1

4

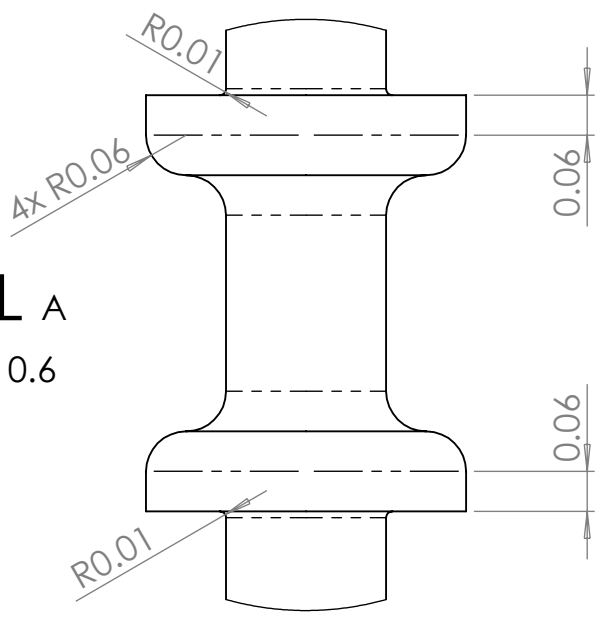
3

2

1



DETAIL A
SCALE 2 : 0.6



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND
BREAK SHARP
EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE		
DRAWN					
CHK'D					
APPV'D					
MFG					
Q.A					

MATERIAL:

WEIGHT:

TITLE:
Main drive shaft for
gear train, Sourceable

DWG NO.

drive_shaft

A4

SCALE:1:0.6

SHEET 1 OF 1